CS177 Bioinformatics: Software Development and Applications

Lecture 3: Algorithms and Probability

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Learning objectives

- At the end of this lecture, you will be able to:
  - Understand what algorithms are
  - Measure running time of an algorithm
  - Understand some common techniques for algorithm design
  - Define and use some basic concepts of probability (e.g. Bayes' theorem, maximum likelihood)

What Is an Algorithm?

Algorithm

- An **algorithm** is a sequence of instructions that one must perform in order to solve a well-formulated **problem**
- Problems are specified in terms of their **inputs** and **outputs**
- The algorithm will be the method of translating the inputs into the outputs
- A well-formulated problem is unambiguous and precise, leaving no room for misinterpretation

Pseudocode

- Using a programming language (e.g. C or Java) to describe an algorithm will contain too many low-level technical details
- **Pseudocode** is a language we often use to describe algorithms. It ignores many details required in a programming language, yet it is more precise and less ambiguous than using a natural language

Variables and arrays

- A **variable**, written as *x* or *total*, contains some numerical value and can be assigned a new numerical value at different points throughout the course of an algorithm
- An **array** of *n* elements is an ordered collection of *n* variables $a_1, a_2, \ldots, a_n$
- Arrays are usually denoted by bold-face letters like $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ and the individual elements are written as a_i, where *i* is between 1 and *n*

Subroutines

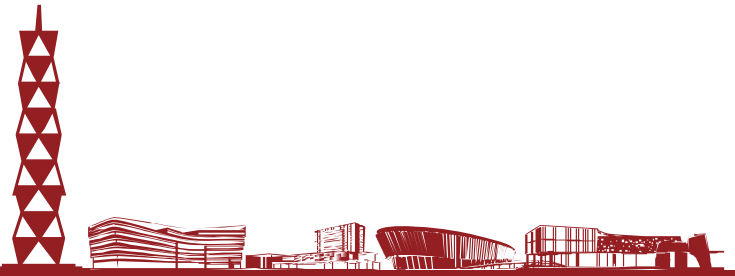
- For clarity and modularity of coding, operations are often grouped into mini-algorithms called **subroutines** (or **functions**)
- An algorithm in pseudocode is denoted by a **name**, followed by a list of **arguments** / **parameters** that it requires, e.g. Max(a, b), which is followed by the statements that describe the algorithm’ s actions
- One can **call** or **invoke** an algorithm by passing it the appropriate values for its arguments
- The operation of **return** reports the results of the algorithm or just signals its end

Basic operations

- Assignment: $a \leftarrow b$
- Arithmetic: $a + b$, $a - b$, $a \cdot b$, a/b, a^b
- Conditional: **if** (A is true) … **else** …
- For loops: **for** i ← a to b {B}
- While loops: **while** (A is true) B
- Array access:
  - a_i denotes the *i* th number of array **a** = $(a_1, a_2, \ldots, a_n)$

Meaning of correct and incorrect algorithms

- We say that an algorithm is **correct**, if it can translate every input instance into the correct output
- An algorithm is **incorrect**, if there is at least one input instance for which the algorithm does not produce the correct output
- A critical but healthy pessimism: Unless you can justify that an algorithm **always** returns correct results, you should consider it to be wrong
- For some difficult problems (e.g. NP-complete problems, to be explained later), however, we rely on **approximation** algorithms, because no practical and correct algorithm can be found



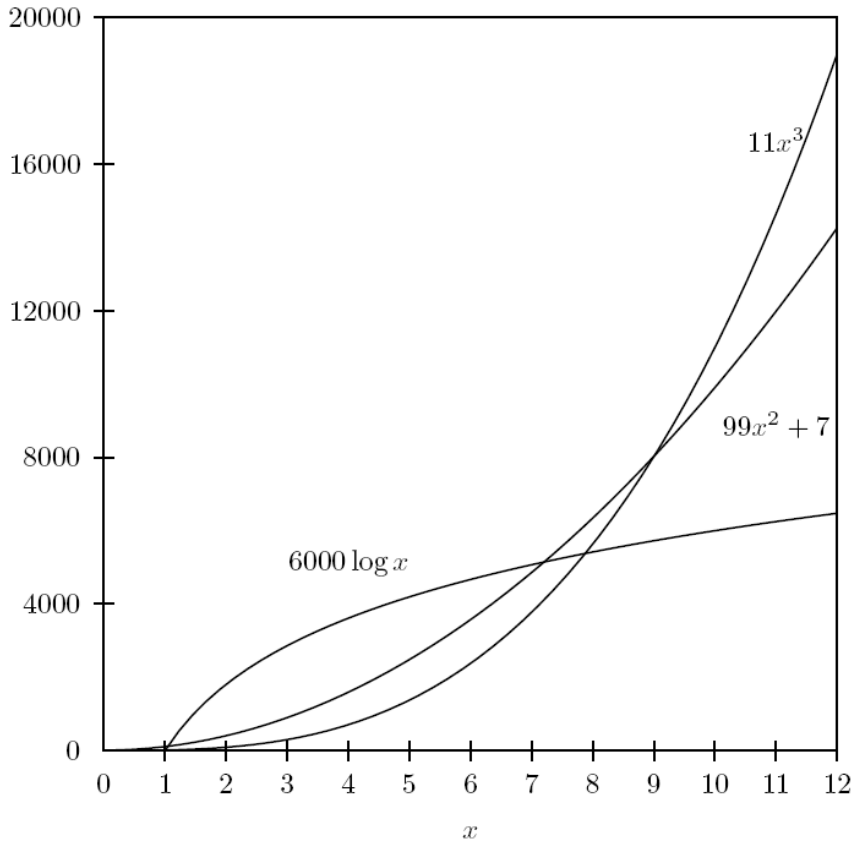
Fast versus Slow Algorithms

Measure of running time

- Suppose our computer takes 10^{-9} second to perform an elementary operation, and we know how many operations an algorithm would perform, then the running time will be the product of the number of operations and the time per operation
- However, an algorithm could be run on different computers. Thus, the time per operation is not fixed
- Hence, rather than computing an algorithm's running time on every computer, we rely on the **total number of operations** that the algorithm performs to describe its running time
- As such, the running time will be an attribute of the algorithm, and not an attribute of a specific computer

Running time as a function of input size

- Typically, the larger the input is, the longer the algorithm takes to process it
- To describe the performance of an algorithm for all kinds of input, and for comparison between algorithms, we need a high-level understanding of the growth of an algorithm's operation count as the input size increases
- Example: To solve a problem on an input of size **n**, algorithm A performs **$11n^3$** operations on an input of size n, and algorithm B performs **$99n^2 + 7$**. Which algorithm is faster?
- For *large n*, B performs much less operations than A, so B is faster (i.e. **more efficient**)



A comparison of a **logarithmic** ($6000\log x$), a **quadratic** ($99x^2+7$) and a **cubic** ($11x^3$) function.

Exponential versus polynomial

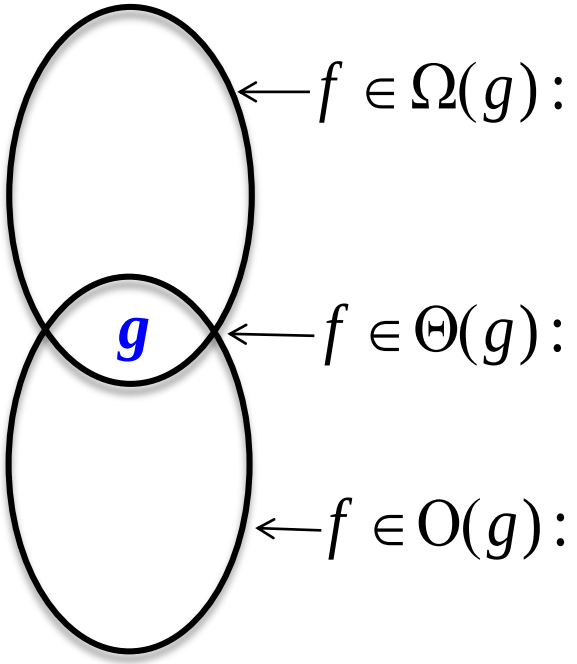
- An algorithm is called an **exponential** algorithm, if its running time includes a term like M^d, where d is a **parameter** of the problem (i.e. d is a **variable** and can be made arbitrarily large by changing the input)
- The running time of a **polynomial** algorithm is bounded by a term like M^k, where k is a **constant** not related to the size of any parameter
- Example: A polynomial algorithm with running time M^1 (linear), M^2 (quadratic), M^3 (cubic), or even M^{2000}

Best, worst and average cases

- **Best-case analysis** gives us the **minimum** number of primitive operations performed by the algorithm on any input of size n (i.e. best-case time complexity)
- **Worst-case analysis** gives us the **maximum** number of primitive operations performed by the algorithm on any input of size n (i.e. worst-case time complexity)
- **Average-case analysis** gives us the **average** number of primitive operations performed by the algorithm on all inputs of size n
- Note:
  - Best case does not mean that n is small (e.g. n = 1), because the running time is represented as a function of n, where n is not fixed
  - For some problems, the input size may include multiple parameters

Big-O notation

- Computer scientists use the Big-O notation to describe the running time of an algorithm **concisely**.
- All functions with a leading terms of n^2 have more or less the same rate of growth, e.g. $99.7n^2$, $0.001n^2$ or $5n^2+3.2n+99993$. We lump them into one class, which we call $O(n^2)$.
- We ignore the leading constant and pay attention only to the **fastest-growing term**.
- Big-Oh (O) , Big-Omega (Ω) and Big-Theta (Θ) are asymptotic (set) notations used for describing the order of growth of a given function.



Algorithm Design Techniques

Problem of searching for a phone

- There are relatively few basic techniques for algorithm design
- To illustrate the techniques, we consider a simple problem:
  - Suppose your mobile phone rings, but you have misplaced it somewhere in your home. How do you find it?
  - To complicate matters, you have just walked into your home with an armful of groceries, and it is dark so you cannot rely solely on eyesight

Exhaustive search

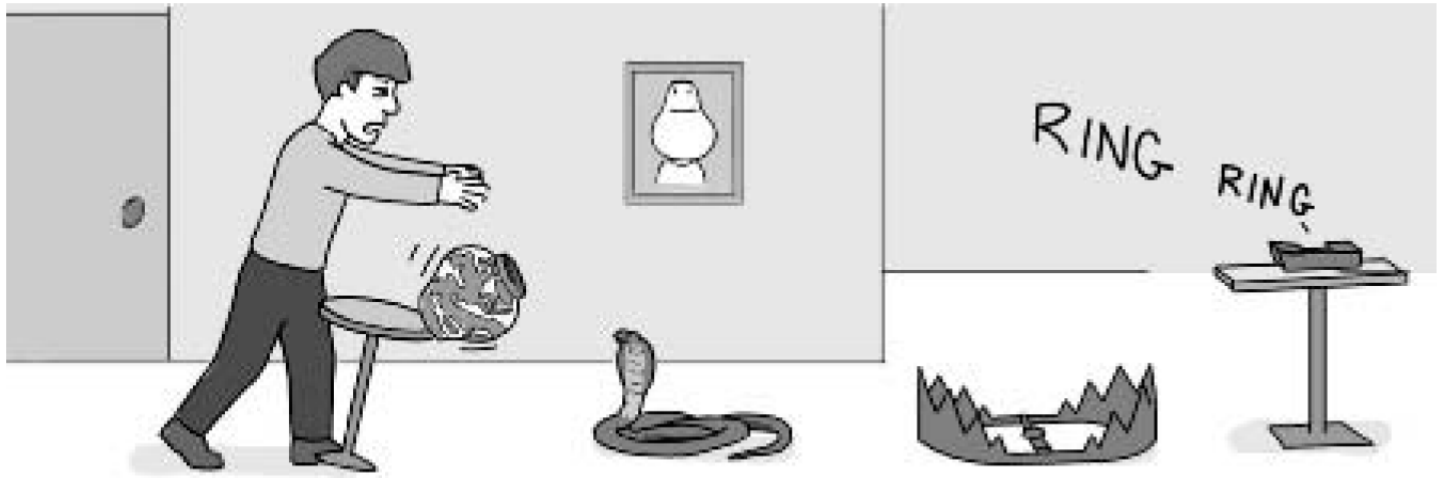
- An **exhaustive search**, or **brute force**, algorithm examines every possible alternative to find one particular solution
- To find the ringing phone, you would ignore the ringing as if you could not hear it, and simply walk over every single square inch of your home
- You would be guaranteed to find the phone no matter where it is
- They are easy to design and understand, but often too slow to be practical

Branch-and-Bound

- As we explore the various alternatives in a brute force algorithm, we may discover that a large number of alternatives can be skipped. This technique is called **branch-and-bound**, or **pruning**
- While exhaustively searching the first floor, if you hear the phone ringing **above** your head, you could immediately rule out the need to search the basement or the first floor
- What might have taken 3 hours may now only take 1 hour

Greedy algorithms

- Many algorithms are iterative procedures that choose among a number of alternatives at each iteration
- **Greedy algorithms** choose the "most attractive" alternative at each iteration.
- Example (US Change problem): The greedy algorithm of BETTERCHANGE chooses the largest denomination possible, i.e. use quarters, then dimes, then nickels, and finally pennies to make change
- Example (Phone searching problem): the greedy algorithm would simply be to walk in the direction of the phone's ringing until you find it. However, there may be a wall (or an expensive and fragile vase) between you and the phone, preventing you from finding it



Divide-and-Conquer

- One big problem may be hard to solve, but two problems of half the size may be much easier
- Steps of a divide-and-conquer algorithm:
  - Split the problem into smaller subproblems
  - Solve the subproblems independently
  - Combine the solutions of the subproblems into a solution of the original problem
- Note: The algorithm usually splits the subproblems into even smaller sub-subproblems, recursively, until it reaches the terminating condition

Dynamic programming

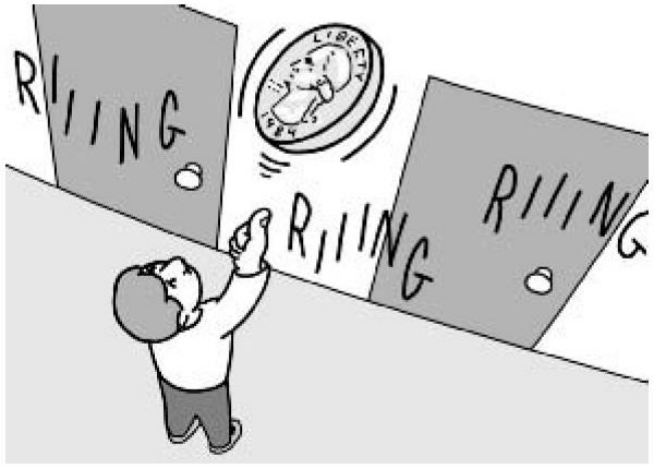
- Some algorithms break a problem into smaller subproblems (e.g. divide-and-conquer). However, the number of subproblems might become very large, and some algorithms solve the same subproblem repeatedly, wasting running time
- Divide-and-conquer is suitable when the subproblems are **independent** of each other. In contrast, dynamic programming is applicable when the subproblems are not independent, that is, when the subproblems share sub-subproblems
- **Rationale of dynamic programming**: Organize computations to avoid recomputing values that you already know, which can save a lot of time
- To avoid recomputing, dynamic programming saves the answer of a sub-subproblem in a **table**, which is looked up when the answer is needed

Machine learning

- A machine learning solution to the phone search problem:
  - Collect statistics over the course of a year about where you leave the phone, learning where the phone tends to end up most of the time
  - Suppose the phone was left in the bathroom 80% of the time, in the bedroom 15% of the time, and in the kitchen 5% of the time
  - Then, a sensible time-saving strategy is to start the search in the bathroom, then the bedroom, and finish in the kitchen
- The performance of a machine learning algorithm often depends on the previously collected data
- The close relation with data analysis makes machine learning methods suitable for bioinformatics

Randomized algorithms

- Before starting to search for the phone, you could toss a coin to decide:
  - If the coin comes up heads, you start the search on the first floor
  - If the coin comes up tails, you start on the second floor
- Sometimes, randomized algorithms do have competitive advantage over deterministic algorithms



Basics of Probability

Sample space and frequency

- **Sample Space** (Ω) of an experiment is the non-empty set of all the mutually exclusive outcomes, e.g.
  - Tossing a coin: Ω = {H, T}
  - Rolling a six-face die: Ω = {1, 2, 3, 4, 5, 6}
  - Randomly selecting a base from DNA sequence: Ω = {A, C, G, T}

- **Frequency of an event** – if an experiment is done n time and an event (the outcome) e occurred $n(e)$ times, then the frequency $f(e)$ of that event occurred in the experiment is

$$f(e) = \frac{n(e)}{n}$$

Probability

- **Events** in nature are deterministic, probabilistic, or fuzzy. The probability is associated with a *stochastic* (*probabilistic*) event.
- If an experiment is performed infinite times, the probability of an event is equal to the frequency of that event:

$$P(e) = \lim_{n \to \infty} \frac{n(e)}{n}$$

- The outcome of a probabilistic event is uncertain and can be stated only with a probability
- A **probability** P of an event e on the sample space Ω is a mapping from Ω to the unit interval [0, 1], $P: \Omega \rightarrow [0, 1]$, which satisfies:
  1. $P(e) \geq 0$;
  2. $P(\Omega) = 1.0$;
  3. $\sum_{e \in \Omega} P(e) = 1.0$ for mutually exclusive and exhaustive events e in the sample space Ω

Conditional probability

- **Joint probability** $P(e_1, e_2)$ is the probability that two (or more) events, say e_1 and e_2, have happened together and is given by the probability of the intersection of two events.
- **Conditional probability** $P(e_1 | e_2)$ of two events, say e_1 given e_2 , is the probability of event e_1 given that the event e_2 (the *prior* event) has already occurred, and is given by

$$P(e_1 | e_2) = \frac{P(e_1, e_2)}{P(e_2)}$$

- Therefore, $P(e_1, e_2) = P(e_1 | e_2) P(e_2)$
- **Independent events**: The probability of one event does not depend on the other event. Therefore, $P(e_1 | e_2) = P(e_1)$ and $P(e_2 | e_1) = P(e_2)$
- That is, the joint probability of independent events is equal to the product of their individual probabilities:

$$P(e_1, e_2) = P(e_1) P(e_2)$$

Bayes' Theorem

- For two events e_1 and e_2, since $P(e_1,e_2)=P(e_1|e_2)P(e_2)=P(e_2|e_1)P(e_1)$,

  Bayes' theorem is given by

$$P(e_1|e_2)=\frac{P(e_2|e_1)P(e_1)}{P(e_2)}$$

- For a given set of mutually exclusive and exhaustive event $e, e_1, e_2, \ldots, e_n$, the probability of event e is give by **the total probability formula (marginalization)**:

$$P(e)=\sum_{i=1}^{n}P(e,e_i)=\sum_{i=1}^{n}P(e|e_i)P(e_i)$$

- For such events, the Bayes' theorem is extended to

$$P(e_i|e)=\frac{P(e|e_i)P(e_i)}{P(e)}=\frac{P(e|e_i)P(e_i)}{\sum_{j=1}^{n}P(e|e_j)P(e_j)}$$

Random variable

- A **random variable** X is defined as a function $X: \Omega \rightarrow D$ on the probability space (Ω, P). The variable is random because the value of the variable is determined probabilistically by P.
- $P(X = x)$ denotes that the probability of the random variable X taking the value x, where $x \in D$.
- Random variables may be:
  - **Discrete**, i.e. the variable is defined on a set of discrete points, or
  - **Continuous**, i.e. the variable is defined on a continuous domain

Probability mass function (pmf)

- For a discrete random variable X taking distinct values, $x_1, x_2, \ldots, x_n$ the **probability mass function**, denoted by p_X, is defined by

$$p_X(x) = \begin{cases} P(X = x_i), & i = 1,2,\ldots,n \\ 0, & \text{otherwise.} \end{cases}$$

and $\sum_{i=1}^{n} p_X(x_i) = 1$

- Notation:
  - Upper-case P denotes a value of probability;
  - Lower-case p denotes a probability distribution;
  - Upper-case X denotes the random variable;
  - Lower-case x denotes the value of random variable.

Probability density function (pdf)

- For a continuous random variable, the probability of a given random variable $X \in \mathbf{R}$ is given by a **probability density function**, $p_X : \mathrm{R} \to [0,1]$ such that

$$P(a \le x \le b) = \int_a^b p_X(x)dx$$

where $a, b \in \mathbf{R}$

- Note: $\int_{-\infty}^{+\infty} p_X(x)dx = 1$ and the probability of a continuous random variable is defined on a **range** of its values but not at **a particular value**

Expectation

- Let X be a random variable and $g: \Omega \rightarrow \mathrm{R}$ be a real-valued function. The **expectation** or **expected value** of the function g with a random variable X, is defined as:

$$E\big(g(X)\big) = \begin{cases} \sum_{x_i \in \Omega} g(x_i) p_X(x_i), & \text{for discrete variables;} \\ \int_{-\infty}^{+\infty} g(x) p_X(x) dx, & \text{for continuous variables.} \end{cases}$$

- **Mean** is the expectation of random variable X, $E[X]$, and is defined as:

$$\mu_X = \begin{cases} \sum_{x_i \in \Omega} x_i p_X(x_i), & \text{for discrete variables;} \\ \int_{-\infty}^{+\infty} x p_X(x) dx, & \text{for continuous variables.} \end{cases}$$

- **Moments** : If X is a random variable, the rth moment of X, denoted by m'_r, is given by $m'_r = E[x^r]$. Note that

$$m'_1 = \mu_X$$

Variance

- **Central moments** : If X is a random variable, the ***r*th central moment** of X **about *a*** is defined as $E[(X-a)^r]$. If $a = \mu_X$, we have the rth central moment about the mean μ_X, which is denoted by m_r :

$$m_r = E[(X - \mu_X)^r]$$

- **Variance** is the 2nd central moment of a variable about its mean, denoted by var[ ]:

$$\text{var}[X] = m_2 = E[(X - \mu_X)^2]$$

- **Standard deviation (SD)**, σ, is the square root of the variance:

$$\sigma_X = (\text{var}[X])^{1/2}$$

Correlation

- **Covariance** : Let X and Y be two random variables defined on the same probability space. Then the **covariance** of X and Y, denoted by cov[X, Y], is defined as

$$\text{cov}[X,Y] = E[(X-\mu_X)(Y-\mu_Y)]$$

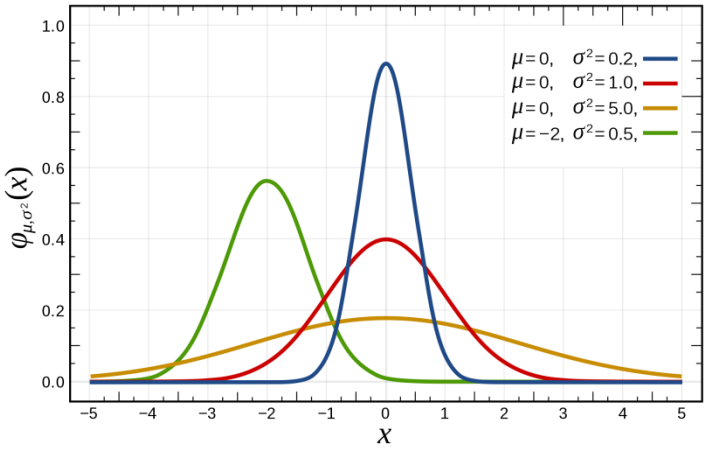
- **Correlation coefficient** is denoted by $\rho[X,Y]$ of two random variables X and Y. It is defined as

$$\rho[X,Y] = \frac{\text{cov}[X,Y]}{\sigma_X \sigma_Y}$$

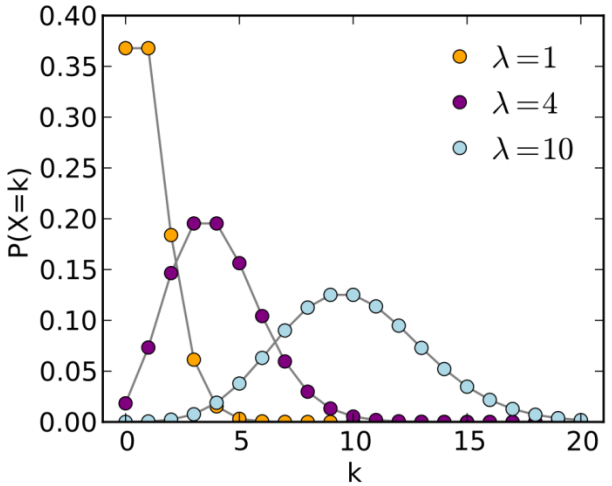
where σ_X and σ_Y are the SD of X and Y, respectively.

Common probability distributions

- Uniform distribution
- Bernoulli distribution
- Binomial distribution
- Multinomial distribution
- Normal (Gaussian) distribution
- Geometric distribution
- Poisson distribution
- Dirichlet distribution
- Gamma distribution
- Chi-square (χ^2) distribution
- etc.



The normal (Gaussian) distribution



The Poisson distribution

Maximum Likelihood

Probabilistic modeling

- Most biological phenomena are due to random events or interpreted with probability
- How to build probabilistic models for biological data, processes, or phenomena?
  - E.g. sequences (DNA, protein, etc.), gene expressions, protein structures, evolution
- Such a model assigns high probability to data/information when it fits the phenomenon well, and low probability for those it does not fit well
- Issues in probabilistic modeling:
  - What is the best model? Multiple models; methods to assess how well a given dataset D fits a model instance M; model selection problem
  - What is the learning algorithm? That is, how to determine the parameters of the model?
  - Amount of data available for model training (or parameter estimation)?
  - Characteristics of data (noise, independence, redundancies, biases)?
  - Unless stated otherwise, the data $d \in D$ are generally assumed to be independent

Maximum Likelihood (ML)

- Once the model M is chosen, the parameters of the model have to be inferred from the data. This is referred to as **learning** or **training** of the model
- Given the model M and its parameters α, the **likelihood** of data D is given by

$$P(D \mid \alpha, M)$$

- The likelihood indicates how well the model predicts the data
- The **maximum likelihood (ML)** estimator maximizes the likelihood of the data given the model. That is, it finds the optimal set of parameters α^{ML} that maximize the likelihood:

$$\alpha^{\mathrm{ML}} = \arg\max_{\alpha} P(D \mid \alpha, M)$$

- Often the log-likelihood (natural logarithm of the likelihood function) is used for computational efficiency.

Strengths and drawback of ML

- **Consistency**: Maximum likelihood is **consistent** in the sense that the true (unknown) parameter α_0 will also, in the limit of a large amount of data, be the value that maximizes the likelihood, i.e., $\alpha \rightarrow \alpha_0$ as the data size $n \rightarrow \infty$
- Other nice properties of ML:
  - Efficient: needs less data than other estimators to achieve a given performance
  - Invariance to parameter transformation: If θ* is the MLE of θ, then for any function f(θ), the MLE of f(θ) is f(θ*)
- Drawback: When the data are scanty, ML can give poor results
  - Example: In rolling a die, to estimate the probabilities of the 6 faces, θ_1, θ_2, …, θ_6, if we use only 3 different rolls of the die, then the ML estimate is

$$\theta_i = n_i / \sum n_k$$

  - But then, at least 3 of the 6 parameters have values 0, a bad estimator
  - Solution: To incorporate *prior* knowledge (e.g. θ_1, θ_2, …, θ_6 should all be near 1/6)

Maximum a posteriori (MAP)

- Given the data *D* and the model *M*, the **posterior probability** is the probability of parameters, $P(\alpha \mid D, M)$
- From Bayes' theorem:

$$P(\alpha \mid D, M) = \frac{P(D \mid \alpha, M)P(\alpha, M)}{P(D, M)} = \frac{P(D \mid \alpha, M)P(\alpha \mid M)P(M)}{P(D, M)}$$

- Because the parameters α do not depend on the terms $P(M)$ or $P(D, M)$,

$$P(\alpha \mid D, M) \propto P(D \mid \alpha, M)P(\alpha \mid M)$$

- The **maximum a posteriori (MAP)** estimator gives the parameters that maximize the posterior probability of the parameters, i.e. the MAP estimator is given by

$$\alpha^{\mathrm{MAP}} = \arg\max_{\alpha} P(D \mid \alpha, M)P(\alpha \mid M)$$

- The **prior probability** of parameters $P(\alpha \mid M)$ is chosen in some reasonable manner to incorporate *prior* (biological) knowledge (Bayesian statistics)

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Recap

- In this lecture, we have learned:
  - What is an algorithm
  - How to use pseudocode to describe algorithms
  - How to measure the running time of algorithms
  - Common techniques of algorithm design
  - Basics of probability theory and probabilistic modeling